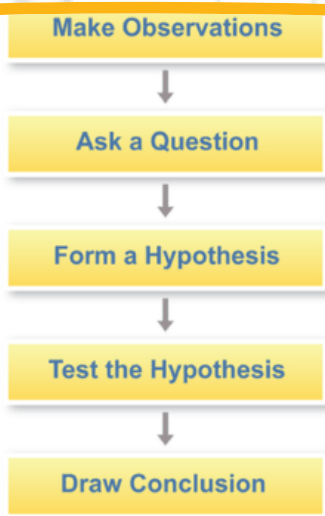




SOLVING A BIOLOGICAL PROBLEM



DEFINITIONS

BIOLOGICAL METHOD:

The scientific method in which biological problems are solved.

THEORY:

It is a set of consistent and experimentally supported scientific assumptions.

DATA:

The information such as name, dates or values made from observation and experimentation.

MATHEMATICAL BIOLOGY:

It is a field of research that examines mathematical representation of biological system.

MATHEMATICAL MODELS:

It is a description of a system using mathematical concept and languages.

SHORT/DETAILED ANSWER & QUESTIONS

Q. Define biological method. Explain in various steps involved in this method

Ans: **BIOLOGICAL METHOD**

Any organized and systematic method which is used to resolve a specific biological problem is called biological method of study.

OR

“The scientific method in which biological problems are solved is called biological method.”

DIFFERENT STEPS OF SOLVING BIOLOGICAL PROBLEMS: In order to resolve a specific biological problem, biologist takes following steps.

1. Observation.
2. Hypothesis.
3. Deduction.
4. Experimentation.
5. Result.
6. Law or Theory.

OBSERVATION:

The first step is to identify the reason of the problem followed by the formulation of a question about what has been observed. In 1880, a French physician, LAVERAN, studied the blood sample of Malaria patient under microscope and observe tiny creatures in it and named as **Plasmodium**.

HYPOTHESIS:

Hypothesis is a key component of the scientific process. It is defined as the “The intelligent guess made by a scientist in the form of statement.” **For example**, in malaria case, an intelligent guess is made after observation that Plasmodium is the cause of malaria.

DEDUCTIONS:

The logical consequences of a hypothesis are called deductions.” For this purpose, a hypothesis is taken as true and expected results are drawn. Generally, in a biological method, if a particular hypothesis is true, then one should expect (deduction) a certain result.

EXPERIMENTS:

Experiments are designed to test the deduction. Biologist perform experiments to see hypothesis as true or not.

RESULT:

The results are where you repeat what happened in the experiment. That includes detailing all observations and data made during your experiment. Result verify the hypothesis.

LAW OR THEORY:

After repeated experimentations and elimination of personal biases the theory makes general principle of law, which is universally accepted, a law is a statement (theory) which is proved by repeated experiments

Theory is made of the hypothesis is experimentally.

“A theory is a statement (hypothesis) which is proved experimentally.”

Q. Explain Biological method with an example of malaria

Ans: **OBSERVATION**

In ancient times, physicians were familiar with this disease. They described it as a fever with recurring attacks, they observed that the disease was more common among people living in low marshy areas.

Major Observations about Malaria:

1. Malaria and marshy areas have some relation.
2. Quinine is an effective drug for treating malaria.
3. Drinking the water of marshes does not cause malaria.
4. Plasmodium is seen in the blood of malarial patients.

HYPOTHESIS:

The hypothesis made regarding malaria was **“Plasmodium is the cause of malaria.”**

DEDUCTION:

Following deduction are drawn from the above-mentioned hypothesis.

“If the plasmodium is the cause of malaria than all persons ill with malaria should have plasmodium in their blood.”

EXPERIMENT:

The next step was to test the deduction through experiments where were designed as.

- Blood of 100 malarial patients was examined under microscope.
- Blood of 100 healthy persons was also examined under microscope.

RESULT:

The result of the experiments shows that almost all malaria patients had plasmodium in their blood while 7 out of 100 healthy persons had plasmodium in their blood. Now we know that plasmodium in the blood of healthy people was in incubation period i.e., the period between the entry of parasite in host and appearance of symptoms. The results were quite convincing and proved that the hypothesis “Plasmodium is the cause of Malaria” was true.

Q. Explain the importance of data organization and data analysis in biological method.

Ans: **Data organization and data analysis are important steps in biological method.**

DATA:

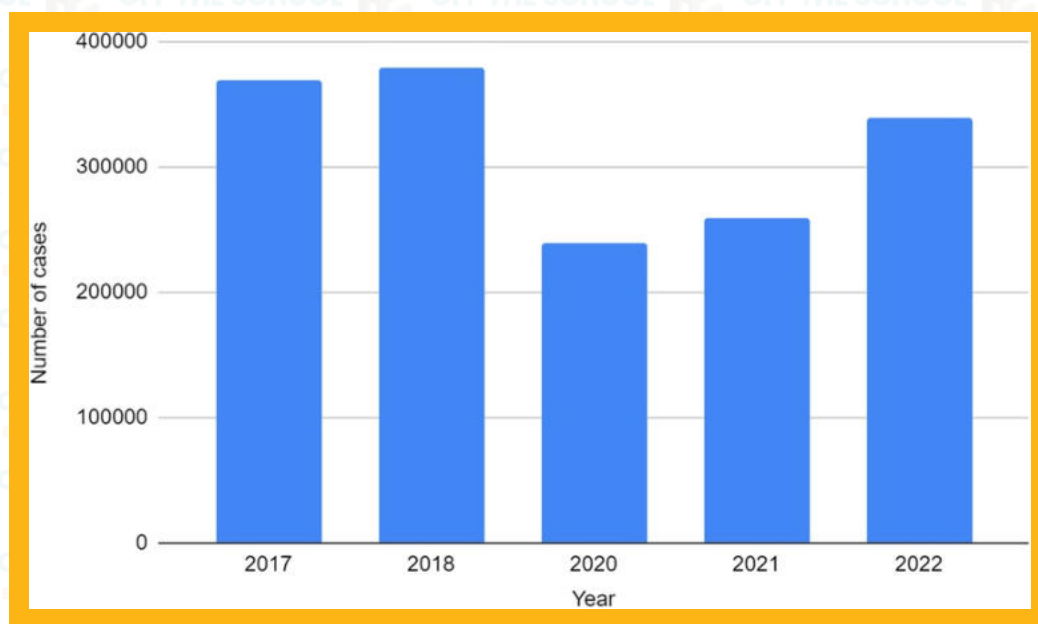
Data can be defined as a single piece of information such as names, dates or values made from observations and experiments.

DATA ORGANIZATION:

In order to formulate and then to test the hypothesis scientist collects and organize data. Prior to conducting an experiment, it is important for a scientist to describe the collection methods because it ensures the quality of experiments. Data is organized in different formats like graphics, tables, flowcharts, maps and diagrams

DATA ANALYSIS:

Data analysis is necessary to prove or disprove a hypothesis by experimentation. It is done through the application of statistical methods.



↑ A chart showing malaria cases reported in Sindh (2017–2022)

Q. Define the following

- Ratio.
- Proportion.
- Mathematical Biology.
- Mathematical Models.

Ans: **RATIO:**

DEFINITION:

When a relation between two numbers e.g., 'a' and 'b' is expressed in terms of quotient (a/b). it is called, the ratio of one number to another.

PROPORTION**DEFINITION:**

Proportion means to join two equal ratios by the sign of equality (=)

EXAMPLE:

A biologist can calculate how many birds could get malaria, when he allowed infected mosquitoes to bite 100 Healthy sparrows. In one experiment, he noted that when he allowed mosquitoes to bite 20 sparrows, 14 out of them got malaria. By using proportion rule:

1st Ratio 14:20 (14 out of 20)
2nd Ratio X:100 (How many out of 100) } Proportion 14:20 :: X:100

$$\frac{X}{100} = \frac{14}{20} \longrightarrow X \times 20 = 100 \times 14 \longrightarrow X = \frac{100}{20} \times 14 \longrightarrow X = 70$$

It means 70 out of 100 Sparrows get malaria

MATHEMATICAL BIOLOGY:**DEFINITION:**

It is a field of research that examines a mathematical representation of a biological systems.

MATHEMATICAL MODELS**DEFINITION:**

One key role of mathematics in biology is the creation of mathematical model

QUALITATIVE OBSERVATIONS	QUANTITATIVE OBSERVATIONS
Qualitative observations are considered less accurate.	Quantitative observations are considered more accurate.
These observations are variable and less measurable.	These observations are in variable and measurable.
These cannot be recorded in terms of numbers.	These can be recorded in terms of numbers.
EXAMPLE: • The freezing point of water is colder than its boiling point. • A liter of water is heavier than a liter of ethanol.	EXAMPLE, • The freezing point of water is 0°C and the boiling point is 100°C. • A liter of water weights 100 gms and a liter ethanol weight 789 gms.

QUALITATIVE OBSERVATIONS	QUANTITATIVE OBSERVATIONS
A hypothesis that stands the test of time often tested and never rejected is called theory.	A scientific law is a uniform, constant fact of nature. It is irrefutable theory.
A theory may be challenged.	A law cannot be challenged.
A theory can be altered in case of new evidence.	Law is already an established and definite entity. It cannot be altered.
A theory is always subjected to new testing.	A law is not subjected to further testing.
EXAMPLE, • Darwin's theory of evolution	EXAMPLE, • Hardy-Weinberg Law

INDUCTIVE REASONING	DEDUCTIVE REASONING
It moves from specific to general.	It moves from general to specific
EXAMPLE, Shark is a fish. All fishes have a scale therefore sharks also have scales.	EXAMPLE, If plasmodium is the cause of malaria, then all the malaria patient should have plasmodium in their blood.

SCIENTIFIC REASONS

Q.1. Theory is highly reliable scientific explanations, why?

Ans. Scientific theories are well tested and highly reliable scientific explanations of natural phenomena that unify many repeated observations and data collected from lots of experiments with authentic resources.

Q.2. Why table or graph is necessary for data organization?

Ans. Data organization requires formation of table or graph for the purpose of mathematical analysis, as the acceptance or rejection of hypothesis is entirely based on the mathematical analysis of recorded data.

Q.3. Why experiment is necessary for theory?

Ans. Experiments plays important role in order to testify theories and to provide basis for scientific knowledge.

